

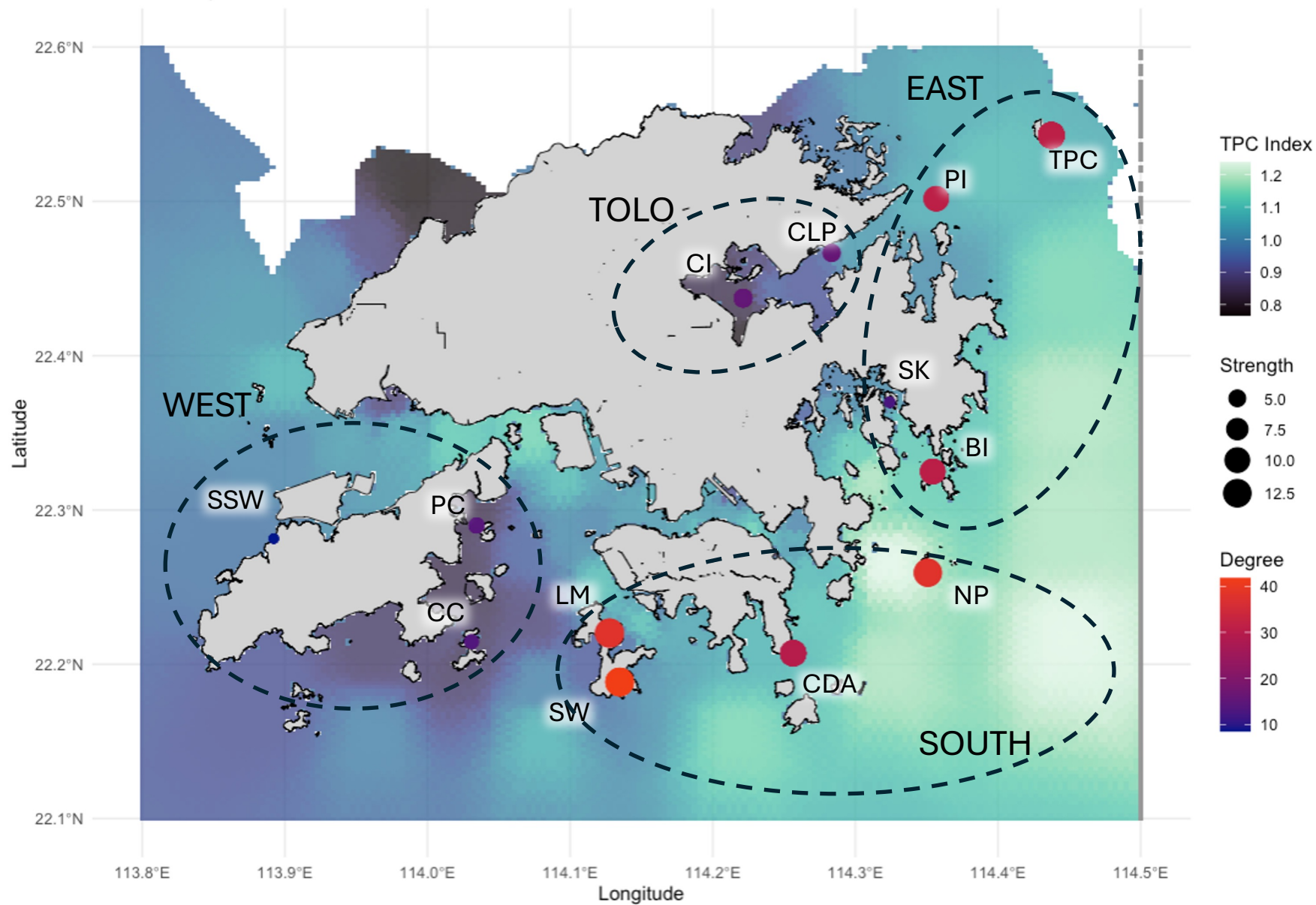


Figure 1: Locations of all 13 sampling sites. Strength and degree were the mean value of all temporal replications. Dotted line outlined the four regions. The background water gradient indicated TPC index modeled with mean Chlorophyll-*a* value between May 2017 – Dec 2022. Since TPC index is linearly modeled by Chlorophyll-*a* concentration, TPC gradient matched Chlorophyll-*a* gradient.